



Evidence Integrity Validator

Forensic File Integrity Validator

User Manual - Version 1.0

For Windows · Works offline · No installation required

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1. Overview

The **Evidence Integrity Validator** is a forensic tool designed for digital investigators, law enforcement professionals, and cybersecurity experts. It allows you to:

- Compute cryptographic hashes of files (MD5, SHA1, SHA256, SHA512, BLAKE2b)
- Extract metadata from images (including GPS coordinates), PDFs, and Office documents
- Verify files against known hash values
- Generate court-ready PDF reports with chain of custody
- Export results to CSV for spreadsheets

Key Benefit: Everything works offline - no internet connection required. Your evidence never leaves your machine.

2. Installation

System Requirements

Operating System	Windows 10 or Windows 11
RAM	512 MB minimum (2 GB recommended)
Disk Space	150 MB for the application
Internet	Not required for operation

Installation Steps

Step 1: Download the `EvidenceValidator.exe` file from the link provided by your distributor.

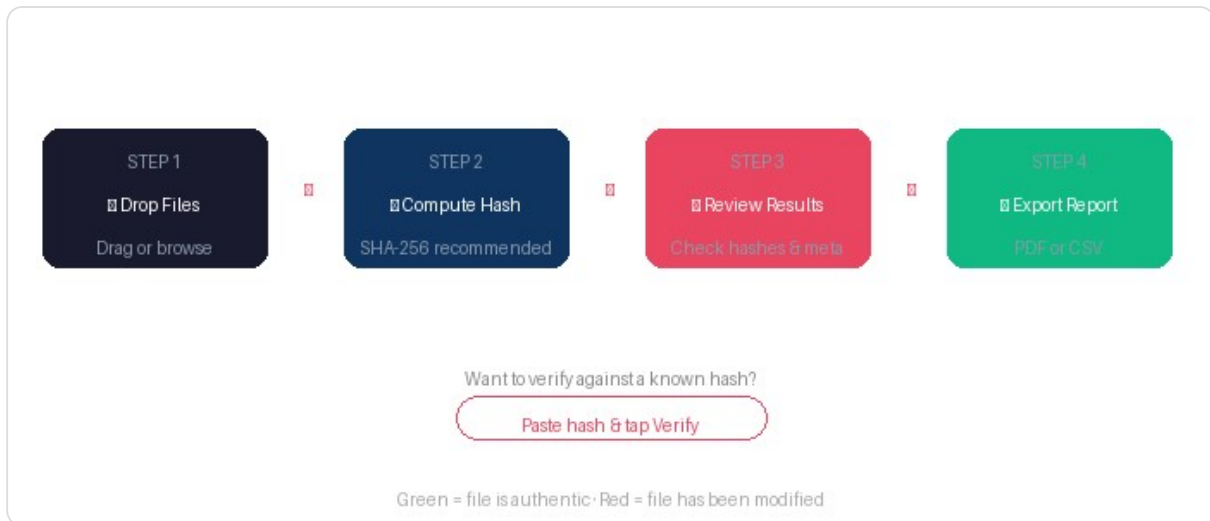
Step 2: Double-click the `.exe` file to run it.

Step 3: If Windows SmartScreen shows a warning, click "**More info**" then "**Run anyway**". This is normal for new software - the program is safe.

Step 4: Your default web browser will open showing the main interface at `http://localhost:8081` .

Note: The application runs as a local web server on your machine only. No data is sent anywhere. Close the terminal window to stop the program.

3. Quick Start Guide



Step 1 - Drop Files: Drag and drop any files into the dashed box on the left panel, or click "browse files" to select them.

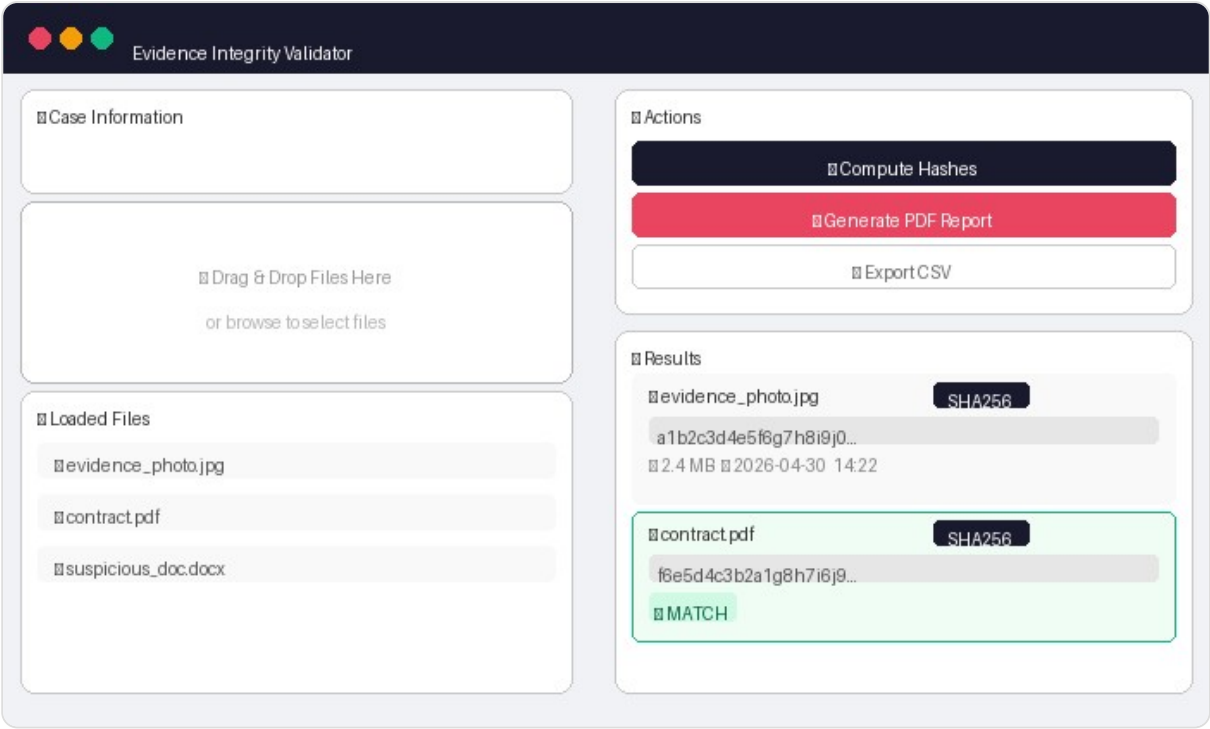
Step 2 - Compute Hashes: Select your hash algorithm (SHA256 is recommended) and click the **"Compute Hashes"** button.

Step 3 - Review Results: Each file shows its hash value, file size, and last modified date. Green badges indicate verified matches.

Step 4 - Generate Report: Click **"Generate Report (PDF)"** to create a court-ready PDF. The report downloads automatically.

Pro Tip: Fill in the Case Information fields first (Case Reference, Examiner Name, Agency) - this information will appear on your PDF report.

4. Features Explained



Main Screen Layout

Area	Description
Case Information	Enter case reference, your name, and agency - printed on reports
Drop Files	Drag & drop area or browse to select files for analysis
Loaded Files	Shows all files added to the current session
Actions	Compute hashes, verify, generate reports, export CSV
Results	Displays computed hashes, file info, and verification status
Metadata	Opens when you click "Metadata" on any result

5. File Hashing

Hashing creates a unique digital fingerprint of a file. Even a single character change in the file produces a completely different hash.

Supported Algorithms

Algorithm	Hash Length	Use Case
MD5	32 characters	Legacy compatibility (not recommended for security)
SHA-1	40 characters	Legacy systems
SHA-256	64 characters	Recommended - Industry standard
SHA-384	96 characters	High-security applications
SHA-512	128 characters	Maximum security
BLAKE2b	128 characters	Fast, modern alternative

How to Hash Files

1. Select your desired algorithm from the dropdown menu (SHA-256 is the default)
2. Ensure your files are loaded in the file list
3. Click "**Compute Hashes**"
4. Results appear in the right panel - each file shows its unique hash
5. Click on any hash value to select it for copying

Why SHA-256? SHA-256 is the industry standard for digital forensics. It's used by law enforcement agencies worldwide and is recognized in court proceedings.

6. Metadata Extraction

Files contain hidden information called metadata. The Evidence Validator can extract this data to help your investigation.

What Can Be Extracted

Images (JPEG, PNG, TIFF)

- Camera make and model
- Date and time taken
- GPS coordinates (latitude/longitude)
- Image dimensions and resolution
- Software used to edit the image
- Flash settings, focal length, ISO

PDF Documents

- Author name
- Creation and modification dates
- Software used to create the PDF
- Number of pages
- Document title and subject

Office Documents (DOCX, XLSX, PPTX)

- Author and last modified by
- Creation and modification dates
- Revision number
- Company name

How to View Metadata

1. Hash your files first (see Section 5)
2. In the results panel, click "**Metadata**" on any file

3. The metadata panel opens below showing all extracted information
4. For images with GPS data, a Google Maps link is automatically generated

7. Hash Verification

Use this feature to verify that a file matches a known hash value - for example, checking a downloaded file against the hash published by the original source.

How to Verify a File

Step 1: Load and hash your file (see Section 5).

Step 2: Paste the expected hash value into the verification box below the "Compute Hashes" button.

Step 3: Click "**Verify**".

The result will show one of two outcomes:

- **MATCH** - The file is authentic and has not been modified
- **MISMATCH** - The file has been altered or corrupted since the original hash was created

Important: The verification tool auto-detects the hash algorithm based on the length of the hash you paste (e.g., 64 characters = SHA-256).

8. Report Generation

The Evidence Validator produces professional forensic reports suitable for submission as evidence in legal proceedings.

What's Included in the PDF Report

- **Case Information:** Case reference, examiner name, agency, date
- **Chain of Custody:** Timestamped record of when files were examined
- **File Details:** Name, path, size, last modified date for each file
- **Hash Values:** Full hash for each file with algorithm specified
- **Verification Results:** Match/mismatch status if verification was performed
- **Extracted Metadata:** All discovered metadata for each file
- **Certification Section:** Signed statement from the examiner

How to Generate a Report

1. Fill in the Case Information fields (Case Reference, Examiner, Agency)
2. Load and hash your files
3. Click "**Generate Report (PDF)**"
4. The PDF downloads automatically

Exporting to CSV

For spreadsheet analysis, click "**Export CSV**" after hashing. This saves the file hashes to a CSV file that can be opened in Excel.

9. Troubleshooting

Application won't start

Problem	Solution
Windows SmartScreen warning	Click "More info" -> "Run anyway"
Nothing happens when double-clicking	Try running as Administrator (right-click -> Run as administrator)
Port 8081 already in use	Close other applications that might use this port

Browser doesn't open

Manually open your browser and go to: `http://localhost:8081`

Large files are slow

Very large files (GB+) take longer to hash. This is normal - the file must be read entirely. Allow time for files over 1GB.

Can't copy hash values

Click directly on the hash text to select it, then press Ctrl+C (Windows) to copy.

Trial limit reached

The trial version allows up to 3 files. To process unlimited files, purchase a license from your distributor.

10. License & Support

Trial Mode

The trial version allows you to process up to **3 files** per session. This lets you fully evaluate the tool before purchasing.

Licensed Version

After purchasing, you'll receive a license key that unlocks:

- Unlimited file processing
- Clean, unwatermarked PDF reports
- Free updates for the lifetime of version 1.x

Contact & Support

To purchase a license or get support:

Email: ibbiy@icloud.com

Best Practices

- Always hash files immediately after acquisition to establish a baseline
- Store hash values separately from the files themselves
- Use SHA-256 as your default algorithm unless a specific standard requires otherwise
- Include chain of custody information in every report
- Keep the PDF reports as part of your case evidence archive